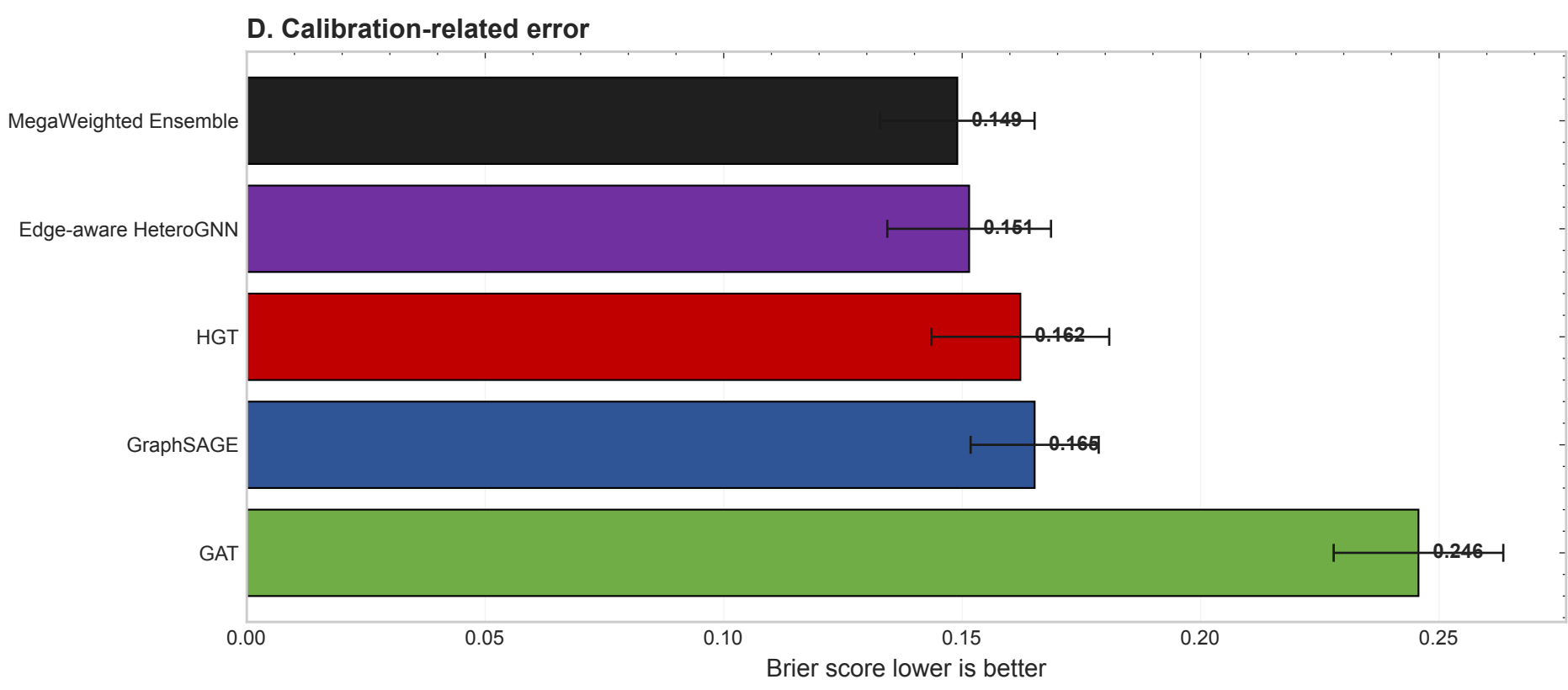
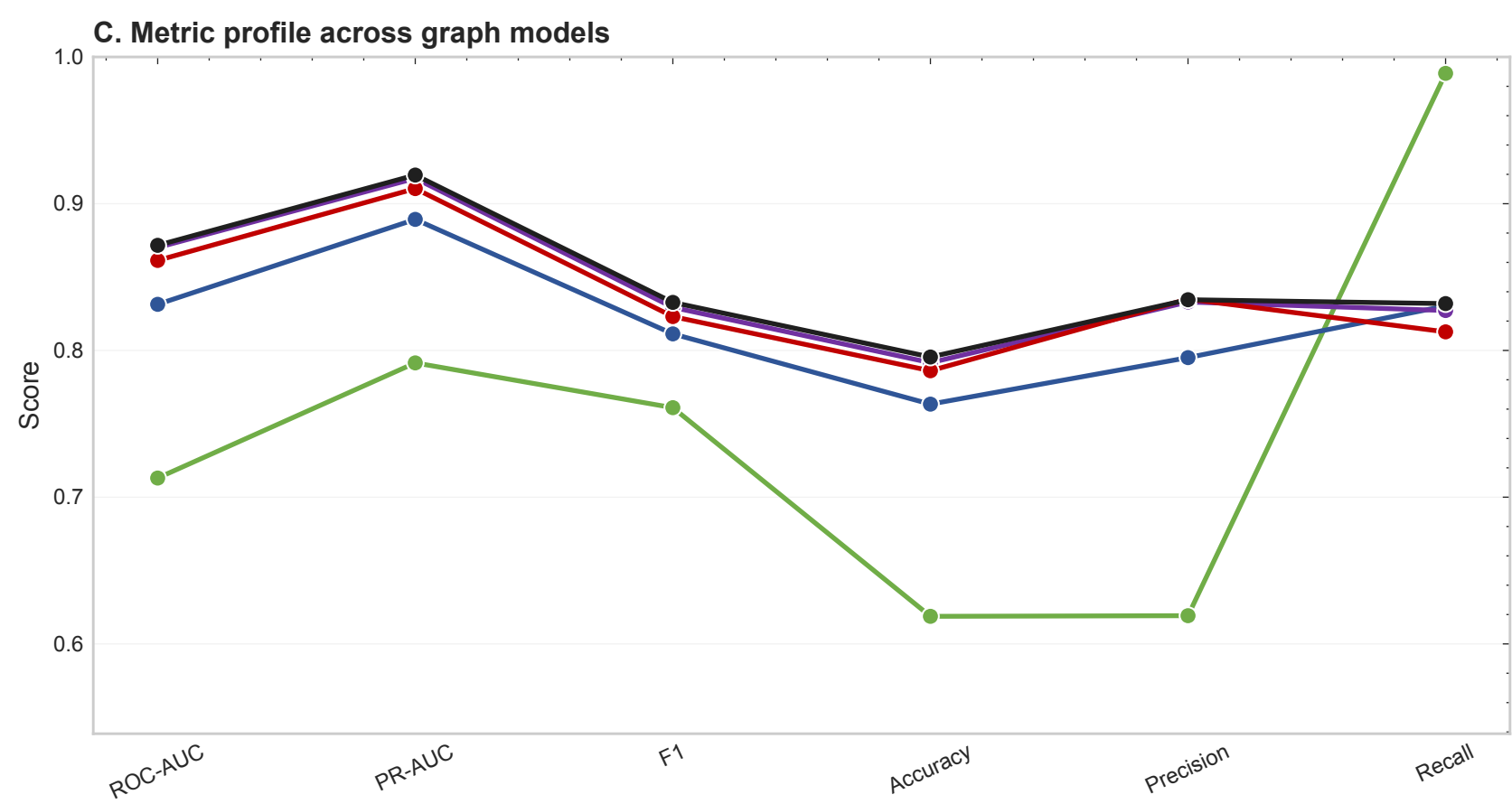
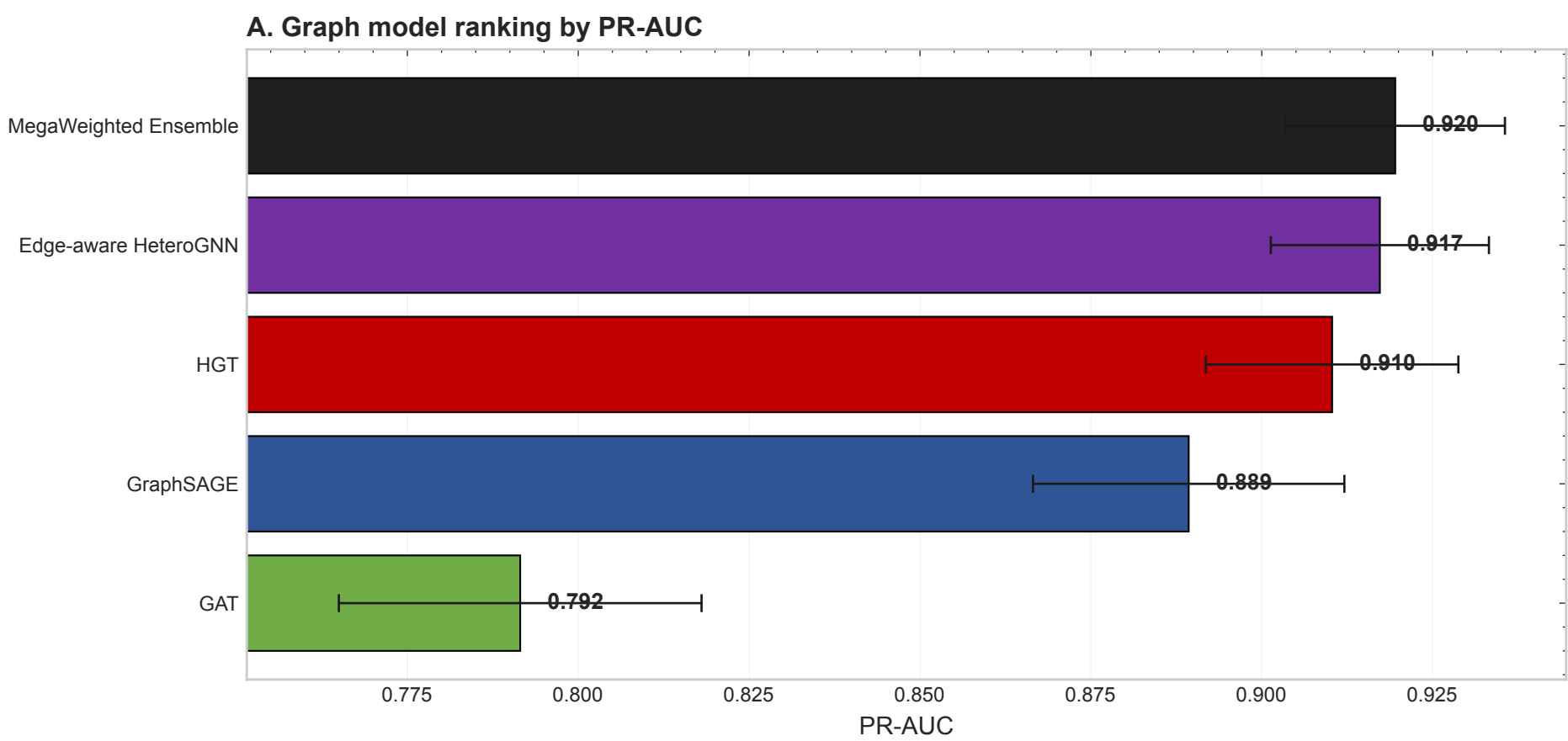
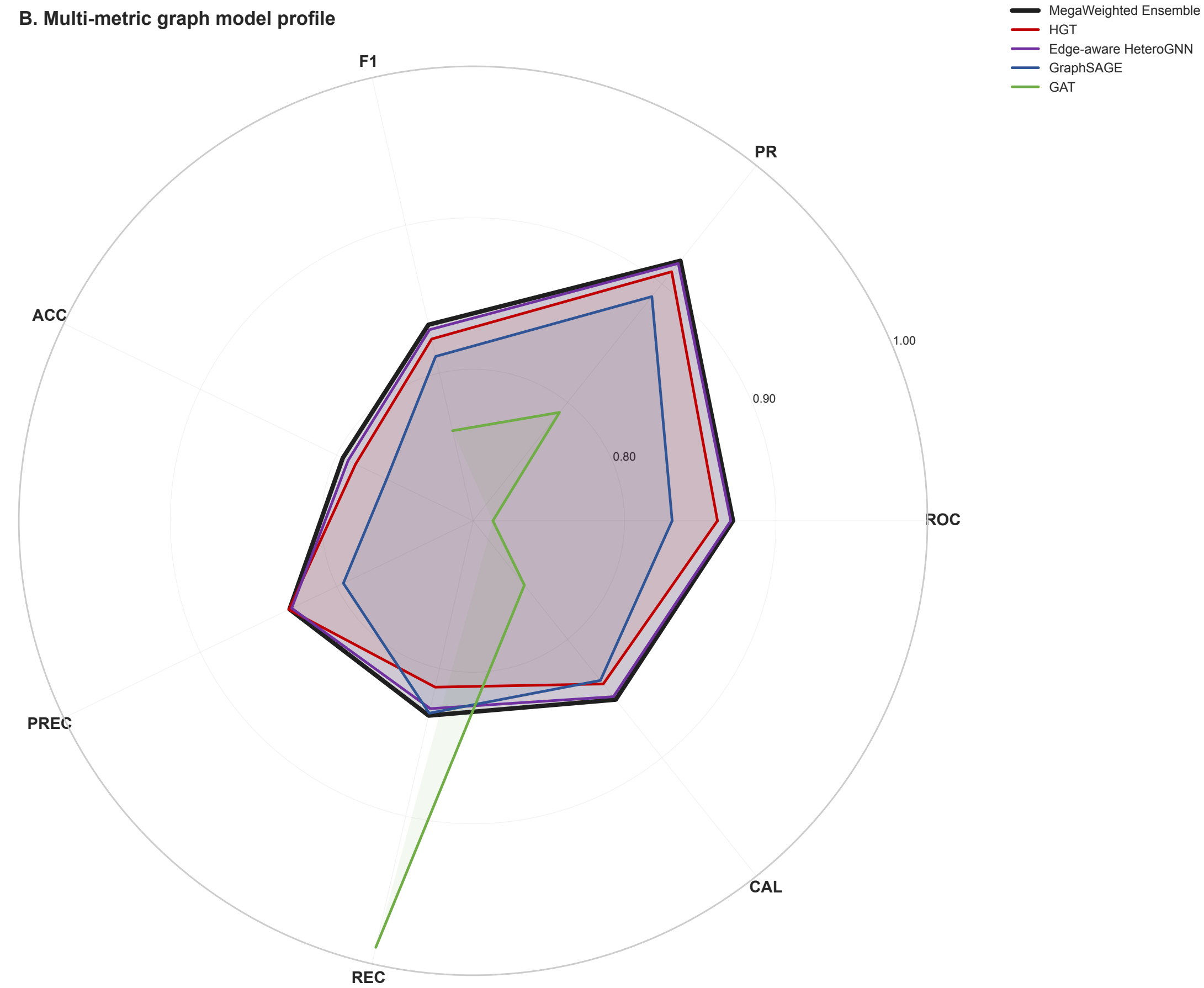


Figure 6. Graph neural network and ensemble benchmark



B. Multi-metric graph model profile



E. Graph model performance summary

Model	ROC	PR	F1	ACC	Brier
GAT	0.713	0.792	0.761	0.619	0.246
GraphSAGE	0.831	0.889	0.811	0.763	0.165
Edge-aware HeteroGNN	0.870	0.917	0.829	0.792	0.151
HGT	0.861	0.910	0.823	0.786	0.162
MegaWeighted Ensemble	0.872	0.920	0.833	0.796	0.149